Using Machine Learning to Predict Customer Engagement in E-commerce.

Farshid Mokhtarinezhad
Faculty of Science
University of Western Ontario
London, Canada
fmokhta@uwo.ca

Yao Li
Faculty of Science
University of Western Ontario
London, Canada
yli4542@uwo.ca

Saad Patel
Faculty of Science
University of Western Ontario
London, Canada
spate696@uwo.ca

***Abstract*—In this project we will be exploring multiple machine learning techniques to predict the shoppers' intentions. For the problem at hand, we have various session information data collected from different visitors of an e-commerce website, the question we are trying to use this data to answer is whether they are going to make a purchase at the end or not. The dataset used in this paper had a large class imbalance, and so oversampling is used to improve the predictions of our models. After exploring AdaBoost, Bagging, Gradient Boosting, KNN, MLP, Random Forest, SVM and Voting classifier we observed that tree based models performed better than the other models.**

***Index Terms*—AdaBoost, Bagging, Gradient Boosting, KNN, MLP, Random Forest, SVM, Voting classifier**

I. Introduction

E-commerce is defined as the activity of electronically buying or selling of products on online services or over the Internet [1]. The rise of ecommerce has greatly shaped the way businesses operate. It has helped businesses establish a wider market presence by providing cheaper and more efficient distribution channels for their products or services [2].

The emergence of analytical tools and predictive algorithms has enabled the ease of doing business for consumers and business owners alike. These tools are especially important for enhancing targeted marketing, considering that visiting an online shopping site does not always necessarily end with a purchase. According to the 2018 Reimagining Commerce Study from Episerver, Only 17% of online shoppers visit shopping sites with the main intention of making a purchase and only about 50% of these customers immediately go directly to the product they are interested in buying [3].

One major area that the business-consumer relationship has been enhanced is through predictive customer preferences during online shopping on e-commerce websites. As research suggests, predictive marketing has evolved from indiscriminate suggestion of products to every user that comes onto an e-commerce site; to a near real time analysis of visitors' information for subsequent follow-up. This system has further transformed into real time analysis of customers and their preferences followed by suggestions of products to avoid the potential loss of business when the customer leaves the e-commerce website (Karim & Mouad , 2020) [4].

Converting intentions into purchases is a huge challenge for businesses. The global retail e-commerce sales in 2020 was - $3.9 trillion [5]. However, if we look at online purchase conversion rates, the average global purchase conversion rate in 2019 was just 2.72% [6].. Moreover, the total online retailing industry loss is $6.5 billion per year due shopping cart abandonment alone [7]. This means it's becoming highly important to access customers 'intentions and preferences using web mining.

A big factor contributing to success in the e-commerce business is understanding how and when customers make purchases online. According to Forbes, the COVID 19 pandemic has shifted the consumer shopping trend towards e-commerce as it saw a significant 129% year-over-year growth in U.S. and Canadian orders as of April 21st 2020, with 146% growth in all online retail orders alone [9]. This means, our customer is now seeking the perfect online shopping experience, re-aligning the direction of the many businesses today towards a new reality.

A powerful application of web mining is in the field of e-commerce as it can be a useful tool in analyzing the effectiveness of marketing campaigns, accessing customer satisfaction and gaining insight into user profiles [9]. This directly enables companies to design their marketing and business strategies and helps them gain competitive advantage. Learning the customer behavior when they are shopping online can increase the profits of a business, by targeting their advertisements towards customers with the same profile as the ones that have purchased from their website, and tailor their listings to keep those customers engaged.

In this project SVM, KNearest Neighbours, Random forest, Bernoulli Naive Bayes, Multilayer Perceptron, AdaBoost, Bagging, Gradient Boosting, and Voting Classifier are used to predict shoppers' intentions. We determined the best parameters for the models by tuning them.

This report is organized in the following sections: Section

2 gives a detailed analysis of the data used for this project. In section 3, an overview of the algorithms used is presented. In section 4, the methodology and approach that was taken to solve this problem is presented. In section 5, the results obtained from applying this methodology is presented. Finally, in section 6, a brief conclusion statement is provided and finally section 7 overlooks some of the future works that can be done to expand and improve this project.

II. Data

The dataset that is being used for this project consists of 18 features belonging to 12,330 sessions, each session belongs to a unique user over a 1-year period to avoid any trends specific to a campaign, special day, user profile, or period. [13]

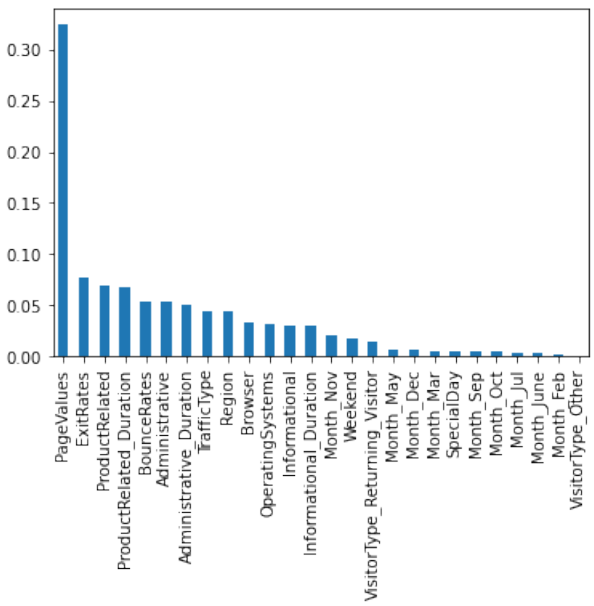

Fig. 1. Attributes of the Datatset

III. Background

A. Algorithms Used

- Support Vector Machines uses a separator as a decision boundary on a small subset of the training data to determine the hyperplane. These data points on the hyperplane are called the support Vectors. It tries to maximize the margin around the separating hyperplane to improve robustness of the network. For non-separable data, it uses a soft margin because this can tolerate errors. Cost finds a balance between making margins large and making sure most points are within the margin. Here the hyperparameter that can be tuned is the value of the cost, the Kernel and the gamma [10].
- In K Nearest Neighbour the distance between test samples with all training data is calculated. Then the k closest entries are chosen and based on the vote of those k neighbours the class of this new sample is specified.
- Random Forest works based on voting among multiple decision trees. Random forest addresses the problem of overfitting in decision tree algorithms.
- Bernoulli Naïve Bayes is based on the concept of conditional probability and falls under the category of Naive Bayes. Naive Bayes is a family of supervised learning algorithms used for classification [11]. It is based on the Bayes theorem which gives the likelihood of an event. A strong assumption about the algorithm that we need to make is that there is a strong (naïve) independence assumption between the features and they do not affect each other's performance. The Bayes theorem describes the probability of an event, based on prior knowledge of conditions that might be related to the event [12]. Bernoulii Naïve Bayes is mostly used when there is discrete data and is based on the Bernoulli distribution. The features mostly used in this case are in binary form, 0 or 1 , true or false.
- A Multilayer Perceptron is a feed-forward neural network composed of at least 3 layers, an input layer, a hidden layer and an output layer.
- AdaBoost which is short for adaptive boosting, fits an estimator on the data and then fits additional copies of the estimator on modified versions of the same data. For each iteration the weights are modified and the model is applied to the re-weighted data. The training examples that were incorrectly classified have their weights increased, and the weights of the correctly classified examples decreased, that way each estimator is forced to focus on the difficult examples misclassified by the previous estimator.
- Bagging an estimator is fit on random subsets of the training set and then their individual predictions are combined to form a final prediction.
- Gradient boosting classifiers are used for classification tasks and consist of two parts, a weak learner and an additive component. They can be defined as a set of machine learning algorithms that combine many weak learning models together to create a strong predictive model using mainly decision trees. The aim of gradient boosting classifiers is to optimize the loss function. [11]
- In a Voting Classifier the final prediction is made based on the majority of the predictions made by each individual classifier in the hopes of increasing the accuracy of the model.

B. Accuracy Measures

- Confusion Matrix is a performance measurement for machine learning classification problems where output can be two or more classes [6]. It can be defined as a table having four combinations of predicted and actual values. The four different combinations are as follows: TP= true positive , FP= false positive, FN= False Negative TN= True Negative.
- Precision is calculated from the confusion matrix as, TP/TP+FP

- Recall is calculated from the confusion matrix as, TP/TP+FN
- F1 score is the weighted average of the precision and recall
- AUROC or the area under the receiver operating characteristic, is a performance metric that can be used to evaluate the model's ability to correctly identify the class of an object. A perfect classifier would have an AUROC score of 1.

IV. Methodology

A. Data Preprocessing

The first step in data processing is data cleaning. In the data at hand there are some entries that are empty. To fill those entries we used the median of the feature. It is common practice to fill in the empty entries with the mean or the median of that feature, if the feature is numerical. We used median as it is a more robust statistical measure and cannot be affected by the outlier points. Also, entries with meaningless data, e.g. time duration with negative entry, are deleted.

In almost all machine learning algorithms, except for tree learning, it is required to have the input with the same range, otherwise some feature would have less effect on the output, while it might not be the case in reality. Thus, it is best practice to normalize all the numerical features before applying any algorithm. Therefore, after cleaning the data the numerical input features were normalized.

The categorical data were then encoded using one-hot encoding. One-hot encoding is one of the most common categorical variable encoding techniques. It adds dummy variables to the data, each of which representing one of the possible values of the corresponding categorical feature. For each sample, all the dummy variables are zero other than the one that represents the value corresponding to the value of the feature in that specific sample.

For the dataset mentioned in section II, there was a large class imbalance, with 7302 false labels and 1445 true labels, the algorithm could predict false every time and it would be accurate 83.5% of the time. To deal with this issue, upsampling was used on the training set to increase the number of true labels to be equal to the false label.

B. Training and Validation process

All the aforementioned algorithms in section III-A were trained once using the original training set, and another time using the upsampled training set. Hyperparameter tuning was performed on the Random Forest, KNN, MLP, and SVM, using 10-fold cross-validation, and the final models were evaluated on the test set.

For the Random Forest model, the number of estimators, the maximum tree depth, the minimum samples required to be at a leaf node, and the maximum number of leaf nodes were tuned. For the SVM model, the regularization parameter C, the degree of the polynomial kernel function, and gamma were tuned. For the KNN model, the number of neighbors to use was tuned. To avoid bias towards one of the training sets, the tuning was performed once with the original training set, and another with the upsampled training set. For the MLP the number of neurons in the hidden layers, the activation function, the solver, the L2 penalty parameter, and the learning rate were tuned.

Boosting techniques work best with weak learners, therefore a decision tree was used as the base estimator for AdaBoost, and Gradient Boosting. On the other hand, Bagging works better with well developed models, therefore the tuned random forest model was used as the base estimator. Random Forest, Gradient Boosting, and AdaBoosting were the classifiers selected to build the Voting classifier.

V. Results

A. Hyperparameter tuning

Table I below shows the range of values used for the hyperparameter tuning of each of the tuned models.

TABLE I
HYPERPARAMETER TUNING

Model	Model Parameter	Range of Values
Random Forest	n_estimators	[10, 100, 1000]
	max_depth	[5,8,15,25,30,None]
	min_samples_leaf	[1,2,5,10,15,100]
	max_leaf_nodes	[2, 5,10]
SVM	n_estimators	[10, 100, 1000]
	C	[0.1, 1, 10, 100, 1000]
	degree	[1-20]
	gamma	[1, 0.1, 0.01, 0.001, 0.0001]
KNN	n_neighbors	[2-10]
MLP	hidden_layer_sizes	[(50,50,50), (50,100,50), (100)]
	activation	[tanh, relu]
	solver	[sgd, adam]
	alpha	[0.0001 ,0.001, 0.05]
	learning_rate	[constant, adaptive]

For the Random Forest model when using the original training set the best parameters were (n_estimators = 10, max_depth = 15, min_samples_leaf = 100, max_lea_nodes = 10). When using the upsampled training set the best parameters were (n_estimators = 1000, max_depth = 8, min_samples_leaf = 10, max_leaf_nodes = 10).

For the SVM model when using the original training set the best parameters were (C = 1000, degree = 18, gamma = 0.001). When using the upsampled training set the best parameters were (C = 1000, degree = 11, gamma = 1).

For the KNN model when using the original training set the best value for K was 9. When using the upsampled training set the best value for K was 2.

For the MLP model when using the original training set the best parameters were found to be (hidden_layer_sizes= (50, 100, 50), activation= tanh, solver= sgd, alpha= 0.05, learning_rate= adaptive). When using the upsampled training set the best parameters were found to be (hidden_layer_sizes= (50, 100, 50), activation= tanh, solver= adam, alpha= 0.0001, learning_rate= adaptive)

B. Random Forest

For the Random forest model tuned and trained using the original training set, an accuracy score of 91% and an AUC score of 0.75 were achieved when evaluated on the test set, in figure 2 and 3 below the confusion matrix and ROC curve for this scenario are presented.

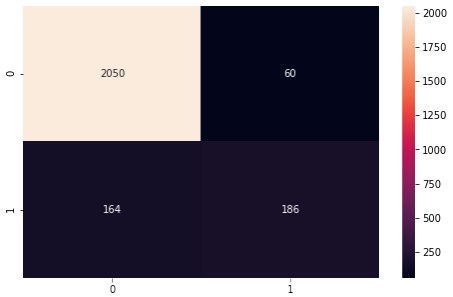

Fig. 2. Confusion matrix for Random Forest trained with unsampled data

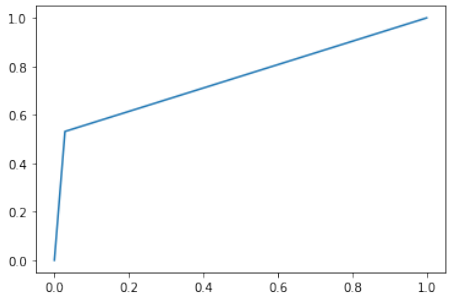

Fig. 3. ROC curve for Random Forest trained with unsampled data

For the Random Forest model tuned and trained using the upsampled training set, an accuracy score of 88% and an AUC score of 0.85 were achieved when evaluated on the test set, in figure 4 and 5 below the confusion matrix and ROC curve for this scenario are presented.

C. SVM

For the SVM model tuned and trained using the original training set, an accuracy score of 90% and an AUC score of 0.72 were achieved when evaluated on the test set, in figure 6 and 7 below the confusion matrix and ROC curve for this scenario are presented.

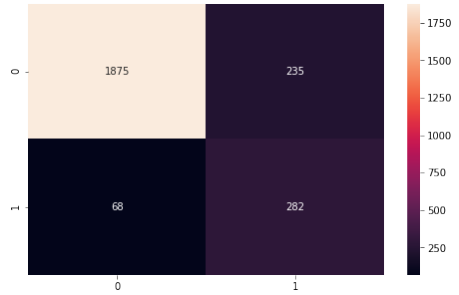

Fig. 4. Confusion matrix for Random Forest trained with upsampled data

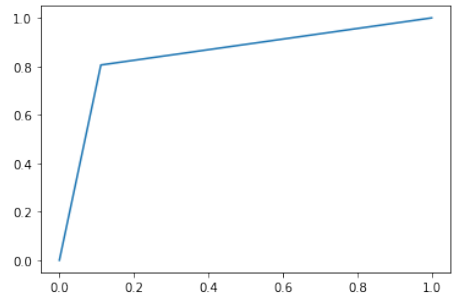

Fig. 5. ROC curve for Random Forest trained with upsampled data

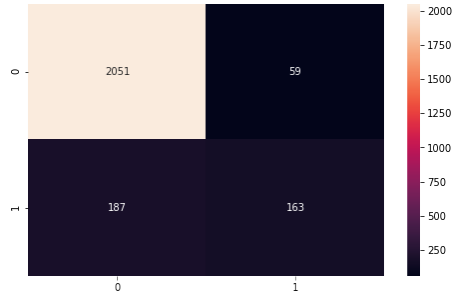

Fig. 6. Confusion matrix for SVM trained with unsampled data

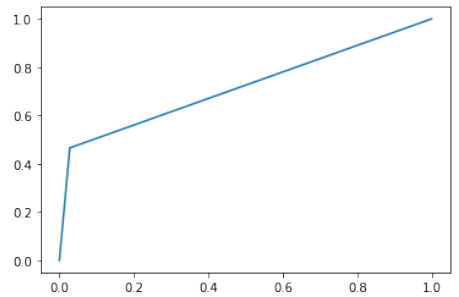

Fig. 7. ROC curve for SVM trained with unsampled data

For the SVM model tuned and trained using the upsampled training set, an accuracy score of 86% and an AUC score of 0.56 were achieved when evaluated on the test set, in figure 8 and 9 below the confusion matrix and ROC curve for this scenario are presented.

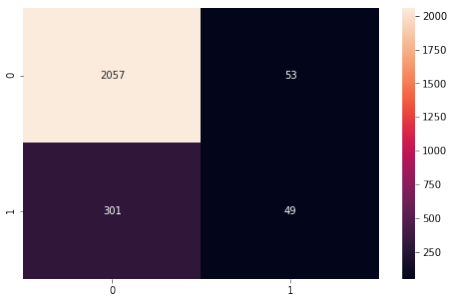

Fig. 8. Confusion matrix for SVM trained with upsampled data

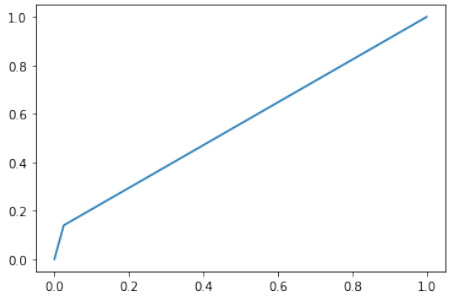

Fig. 9. ROC curve for SVM trained with upsampled data

D. KNN

For the KNN model tuned and trained using the original training set, an accuracy score of 89% and an AUC score of 0.67 were achieved when evaluated on the test set, in figure 10 and 11 below the confusion matrix and ROC curve for this scenario are presented.

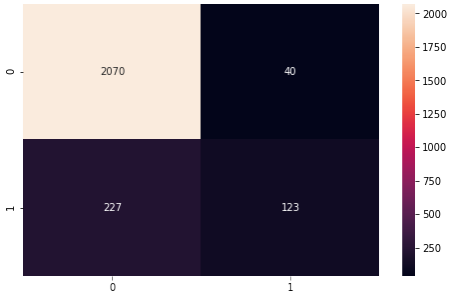

Fig. 10. Confusion matrix for KNN trained with unsampled data

For the KNN model tuned and trained using the upsampled training set, an accuracy score of 83% and an AUC score of 0.69 were achieved when evaluated on the test set, in figure 12 and 13 below the confusion matrix and ROC curve for this scenario are presented.

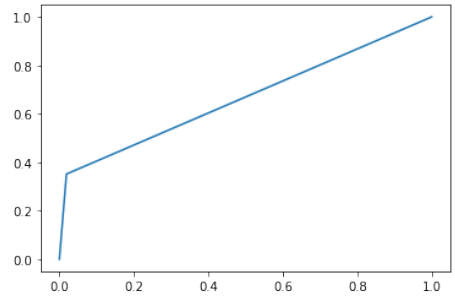

Fig. 11. ROC curve for KNN trained with unsampled data

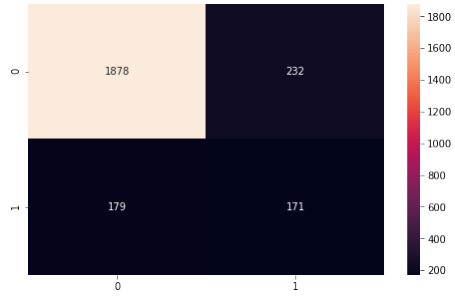

Fig. 12. Confusion matrix for KNN trained with upsampled data

E. Bernoulli Naïve Bayes

For the Bernoulli Naïve Bayes model trained using the original training set, an accuracy score of 85% and an AUC score of 0.74 were achieved when evaluated on the test set, in figure 14 and 15 below the confusion matrix and ROC curve for this scenario are presented.

For the Bernoulli Naïve Bayes model trained using the upsampled training set, an accuracy score of 78% and an AUC score of 0.79 were achieved when evaluated on the test set, in figure 16 and 17 below the confusion matrix and ROC curve for this scenario are presented.

F. Gradient Boosting

For the Gradient Boosting model trained using the original training set, an accuracy score of 92% and an AUC score of 0.79 were achieved when evaluated on the test set, in figure

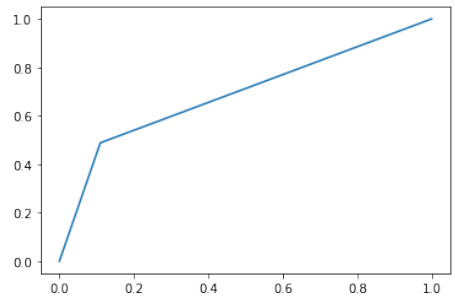

Fig. 13. ROC curve for KNN trained with upsampled data

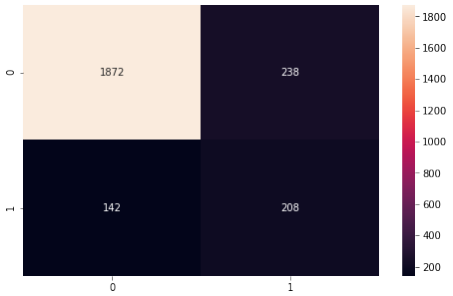

Fig. 14. Confusion matrix for Bernoulli Naïve Bayes trained with unsampled data

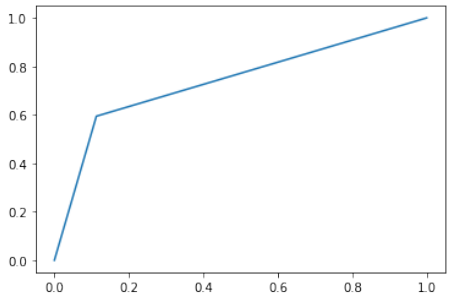

Fig. 15. ROC curve for Bernoulli Naïve Bayes trained with unsampled data

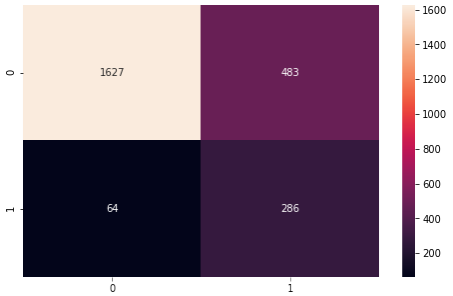

Fig. 16. Confusion matrix for Bernoulli Naïve Bayes trained with upsampled data

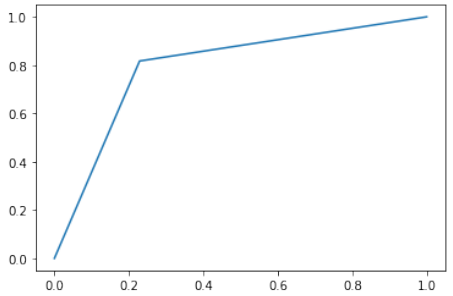

Fig. 17. ROC curve for Bernoulli Naïve Bayes trained with upsampled data

18 and 19 below the confusion matrix and ROC curve for this scenario are presented.

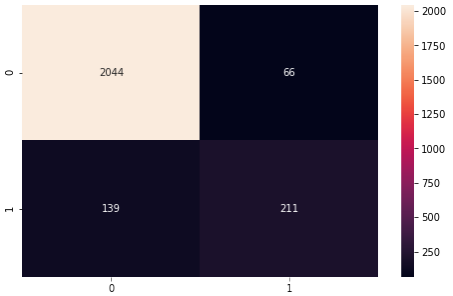

Fig. 18. Confusion matrix for Gradient Boosting trained with unsampled data

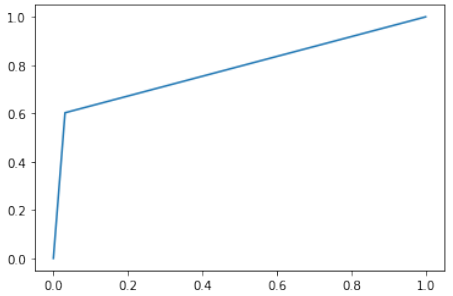

Fig. 19. ROC curve for Gradient Boosting trained with unsampled data

For the Gradient Boosting model trained using the upsampled training set, an accuracy score of 90% and an AUC score of 0.83 were achieved when evaluated on the test set, in figure 20 and 21 below the confusion matrix and ROC curve for this scenario are presented.

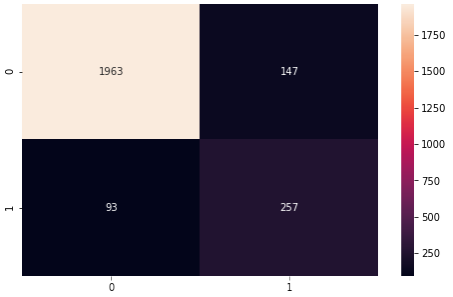

Fig. 20. Confusion matrix for Gradient Boosting trained with upsampled data

G. Bagging

For the Bagging model trained using the original training set, an accuracy score of 92% and an AUC score of 0.76 were achieved when evaluated on the test set, in figure 22 and 23 below the confusion matrix and ROC curve for this scenario are presented.

For the Bagging model trained using the upsampled training set, an accuracy score of 88% and an AUC score of 0.85 were achieved when evaluated on the test set, in figure 24 and 25

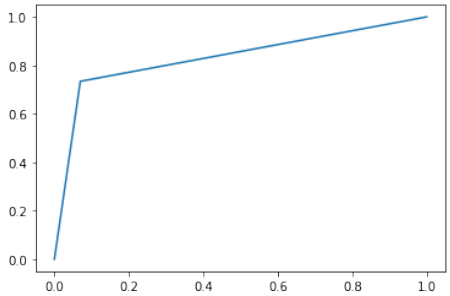

Fig. 21. ROC curve for Gradient Boosting trained with upsampled data

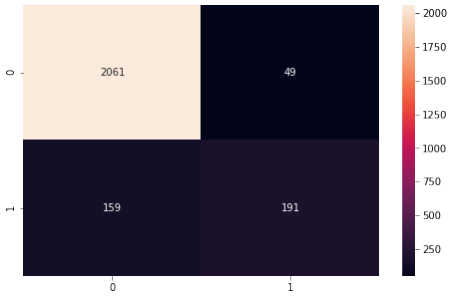

Fig. 22. Confusion matrix for Bagging trained with unsampled data

below the confusion matrix and ROC curve for this scenario are presented.

H. AdaBoost

For the AdaBoost model trained using the original training set, an accuracy score of 90% and an AUC score of 0.76 were achieved when evaluated on the test set, in figure 26 and 27 below the confusion matrix and ROC curve for this scenario are presented.

For the Gradient Boosting model trained using the upsampled training set, an accuracy score of 89% and an AUC score of 0.82 were achieved when evaluated on the test set, in figure 28 and 29 below the confusion matrix and ROC curve for this scenario are presented.

I. Voting Classifier

For the Voting Classifier model trained using the original training set, an accuracy score of 92% and an AUC score of

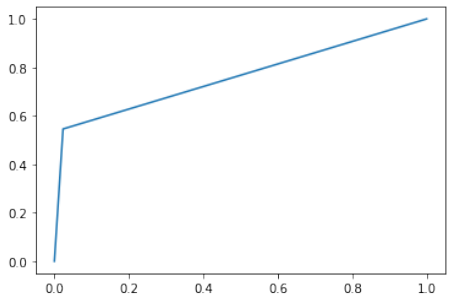

Fig. 23. ROC curve for Bagging trained with unsampled data

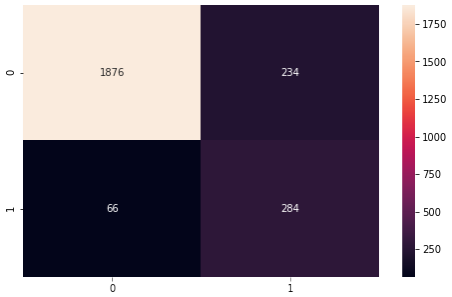

Fig. 24. Confusion matrix for Bagging trained with upsampled data

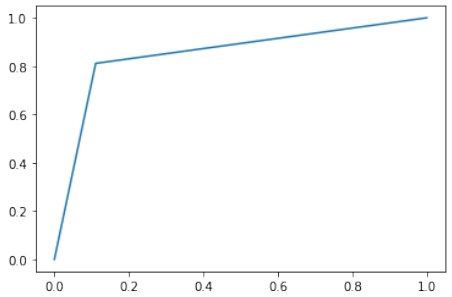

Fig. 25. ROC curve for Bagging trained with upsampled data

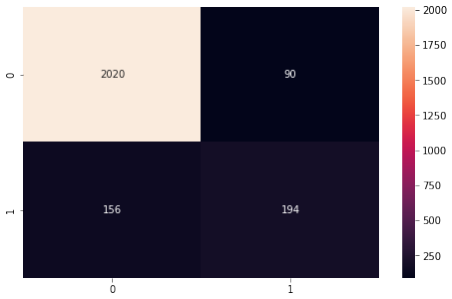

Fig. 26. Confusion matrix for AdaBoost trained with unsampled data

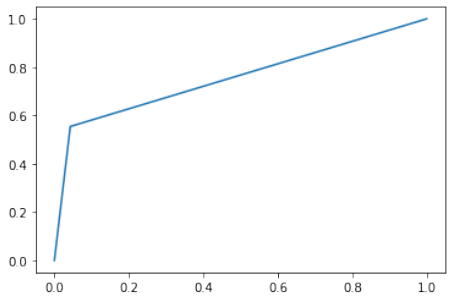

Fig. 27. ROC curve for AdaBoost trained with unsampled data

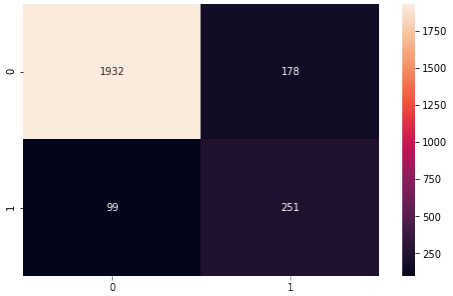

Fig. 28. Confusion matrix for AdaBoost trained with upsampled data

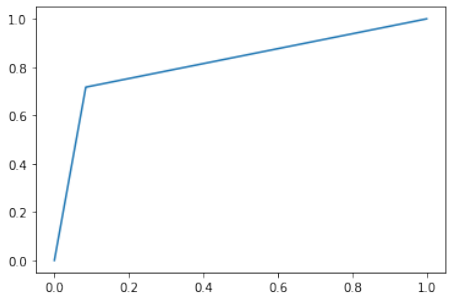

Fig. 29. ROC curve for AdaBoost trained with upsampled data

0.77 were achieved when evaluated on the test set, in figure 30 and 31 below the confusion matrix and ROC curve for this scenario are presented.

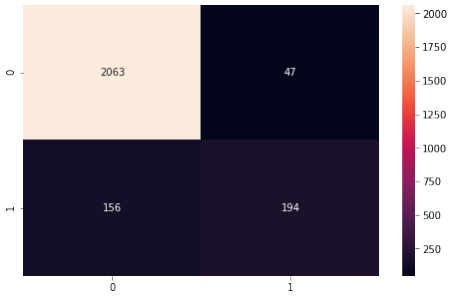

Fig. 30. Confusion matrix for Voting Classifier trained with unsampled data

For the Voting Classifier model trained using the upsampled training set, an accuracy score of 89% and an AUC score of 0.84 were achieved when evaluated on the test set, in figure 32 and 33 below the confusion matrix and ROC curve for this scenario are presented.

J. MLP

For the MLP model trained using the original training set, an accuracy score of 91% and an AUC score of 0.78 were achieved when evaluated on the test set, in figure 34 and 35 below the confusion matrix and ROC curve for this scenario are presented.

For the MLP model trained using the upsampled training set, an accuracy score of 86% and an AUC score of 0.75 were achieved when evaluated on the test set, in figure 36 and 37

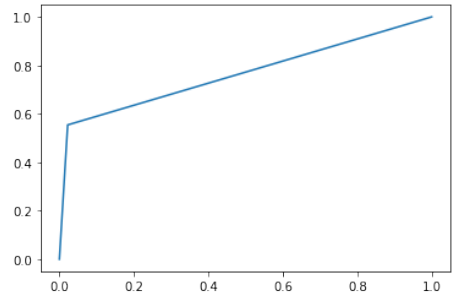

Fig. 31. ROC curve for Voting Classifier trained with unsampled data

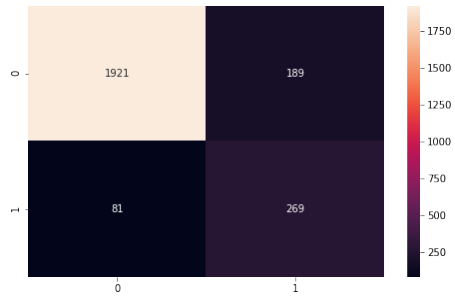

Fig. 32. Confusion matrix for Voting Classifier trained with upsampled data

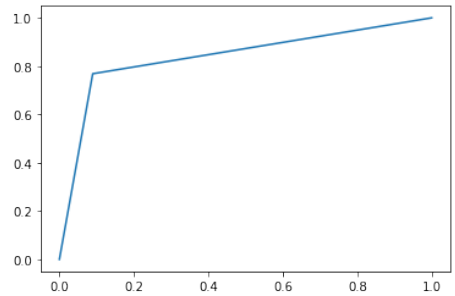

Fig. 33. ROC curve for Voting Classifier trained with upsampled data

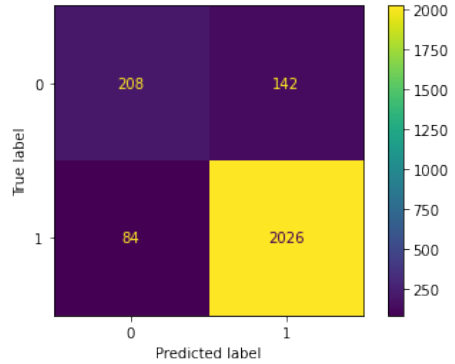

Fig. 34. Confusion matrix for MLP trained with unsampled data

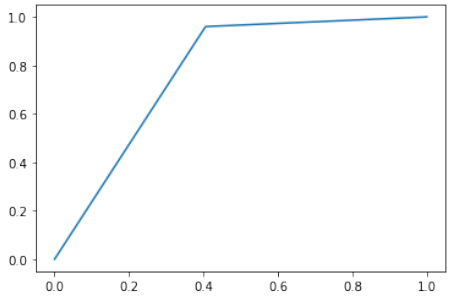

Fig. 35. ROC curve for MLP trained with unsampled data

below the confusion matrix and ROC curve for this scenario are presented.

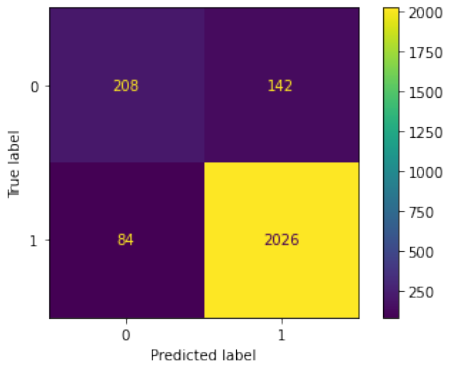

Fig. 36. Confusion matrix for MLP trained with upsampled data

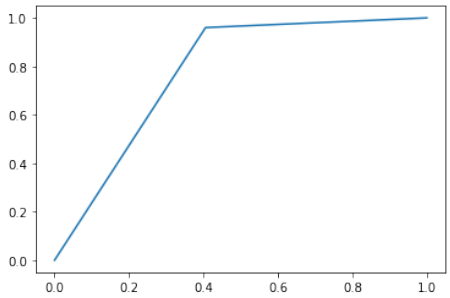

Fig. 37. ROC curve for MLP trained with upsampled data

K. Results

Table III below presents a summary of results for the top 5 algorithms when trained using the upsampled training data.

VI. Conclusion and Future Works

In this project, we explored the effectiveness of different algorithms in predicting whether a shopper is going to generate revenue for a business or not based on their session information. We also experimented with upsampling to observe its impact on the performance of the algorithms when dealing with an imbalanced dataset. We observed that a bagging classifier with tuned Random Forest model as its base estimator performed better than the other algorithms when classifying the customers that generated revenue for the business. We also observed that upsampling increased the recall and AUROC for all the models trained for this paper, except for the MLP, for which these metrics dropped.

TABLE II
Upsampled Training Data Results

Classifier	Accuracy	Precision	Recall	F1	AUROC
Random Forest	88%	0.55	0.81	0.65	0.85
Gradient Boosting	90%	0.64	0.73	0.68	0.83
AdaBoost	89%	0.59	0.72	0.64	0.82
Bagging	89%	0.55	0.81	0.65	0.85
Voting	90%	0.59	0.77	0.67	0.84

TABLE III
Upsampled Training Data Results

Classifier	Accuracy	Precision	Recall	F1	AUROC
Random Forest	91%	0.76	0.53	0.62	0.75
Gradient Boosting	92%	0.76	0.60	0.67	0.79
AdaBoost	90%	0.68	0.55	0.61	0.76
Bagging	92%	0.80	0.55	0.65	0.76
Voting	92%	0.80	0.55	0.66	0.77

In the future, we can try to group similar traffic types of customers together in the preprocessing part to avoid so many dummy variables for the models. So as the following variables: operating systems, browser, and region. We can preprocess them in a more delicate way. These would help achieve our goal of having a more accurate prediction about if the customer will purchase or not. We can also check if there is a seasonality effect in the purchase data, which can help us understand the customer behaviors better.

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